HW #8

3.17

$$\frac{d}{dt}\langle\hat{\varphi}\rangle = \frac{i}{\hbar}\langle[\hat{H},\hat{\varphi}]\rangle + \langle\frac{\partial}{\partial t}\hat{\varphi}\rangle$$

In all cases to be considered: $\langle\frac{\partial}{\partial t}\hat{\varphi}\rangle = 0$ because the operators do not have explicit time dependence

(a) $\frac{d}{dt}\langle\hat{\varphi}\rangle \rightarrow \hat{\varphi} = 1$

$$\frac{d}{dt}\langle 1\rangle = \frac{d}{dt}\int\psi^*\psi\,dt = \frac{i}{\hbar}\langle[\hat{H},1]\rangle = \frac{i}{\hbar}\langle(H-H)\rangle = 0$$

$\Rightarrow$ Constant Normalization with time.

(b) $\hat{\varphi} = \hat{H} \Rightarrow \underbrace{\frac{d}{dt}\langle\hat{H}\rangle} = \frac{i}{\hbar}\langle[\hat{H},\hat{H}]\rangle = \frac{i}{\hbar}\langle\hat{H}\hat{H}-\hat{H}\hat{H}\rangle = 0$

$\langle\hat{H}\rangle \equiv$ energy expectation value

$\frac{d\langle\hat{H}\rangle}{dt} = 0 \Rightarrow$ Energy expectation value remains constant with time

(c) $\hat{\varphi} = \hat{x} \Rightarrow \frac{d}{dt}\langle\hat{x}\rangle = \frac{i}{\hbar}\langle[\hat{H},\hat{x}]\rangle = \frac{i}{\hbar}\langle[\frac{\hat{p}^2}{2m}+\hat{V}(x),\hat{x}\rangle]$

$$= \frac{i}{\hbar}\langle[\hat{p}^2/2m,\hat{x}]\rangle + \frac{i}{\hbar}\langle\cancelto{0}{[V(\hat{x}),\hat{x}]}\rangle$$

position operators commute

$$= \frac{i}{2m\hbar}\langle\hat{p}\underbrace{[\hat{p},\hat{x}]}_{-i\hbar} + \underbrace{[\hat{p},\hat{x}]}_{-i\hbar}\hat{p}\rangle;$$

use: Eqn. 3.64 $[AB,C] = A[B,C]+[A,C]B$ ($A = \hat{p}$, $B = \hat{p}$, $C = \hat{x}$)

$$= \frac{i}{2m\hbar}\langle(-2i\hbar\hat{p})\rangle = \frac{\langle\hat{p}\rangle}{m};$$ Ehrenfest's thm holds!

(d) $\hat{\varphi} \equiv \hat{p} \Rightarrow \frac{d}{dt}\langle\hat{p}\rangle = \frac{i}{\hbar}\langle[\hat{H},\hat{p}]\rangle = \frac{i}{\hbar}\langle[\frac{\hat{p}^2}{2m}+V(x),\hat{p}]\rangle = \frac{i}{\hbar}\langle\cancelto{0}{[\frac{\hat{p}^2}{2m},\hat{p}]}\rangle + \frac{i}{\hbar}\langle[V(x),\hat{p}]\rangle$

and $(V(x)\hat{p}-\hat{p}V(x))\psi \equiv V(x)(-i\hbar\frac{\partial}{\partial x}\psi) - (-i\hbar\frac{\partial}{\partial x}(V\psi)) = +i\hbar(\frac{\partial V}{\partial x})\psi \Rightarrow [V(x),\hat{p}] = +i\hbar\frac{\partial V}{\partial x}$

$$\Rightarrow \quad \frac{d}{dt}\langle \hat{p} \rangle = \frac{i}{\hbar}\left\langle +i\hbar \frac{\partial V}{\partial x} \right\rangle = -\left\langle \frac{\partial V}{\partial x} \right\rangle \longrightarrow$$ Ehrenfest's theorem holds

P 3.22

(a)

$$|\alpha\rangle = i|1\rangle - 2|2\rangle - i|3\rangle$$

$$|\beta\rangle = i|1\rangle + 2|3\rangle$$

transpose

$$\langle\alpha| = \left[(|\alpha\rangle)^*\right]^T = -i\langle 1| - 2\langle 2| + i\langle 3|$$

$$\langle\beta| = \left[(|\beta\rangle)^*\right]^T = -i\langle 1| + 2\langle 3|$$

(b)

$$\langle\alpha|\beta\rangle = 1 + 2i$$

$$\langle\beta|\alpha\rangle = 1 - 2i = \left[\langle\alpha|\beta\rangle\right]^*$$

(c)

$$\hat{A} \equiv |\alpha\rangle\langle\beta|$$

$$A_{ij} \equiv \langle i|\hat{A}|j\rangle \; ; \quad \text{For example} \rightarrow \langle 1|\hat{A}|3\rangle = A_{13} = \langle 1|\alpha\rangle\langle\beta|3\rangle$$

i^{th} row, j^{th} column

1st row, 3rd column

$$= 2i$$

$$\langle 3|\hat{A}|2\rangle = A_{32} = \langle 3|\alpha\rangle\underbrace{\langle\beta|2\rangle}_{0}$$

$$= 0$$

$$A \equiv \begin{pmatrix} 1 & 0 & 2i \\ 2i & 0 & -4 \\ -1 & 0 & -2i \end{pmatrix}$$

$$(A^*)^T = \begin{pmatrix} 1 & -2i & -1 \\ 0 & 0 & 0 \\ -2i & -4 & 2i \end{pmatrix} \; ; \quad \text{since } (A^*)^T \neq A$$

Not Hermitian

3.27

$$\hat{A}\,\psi_1 = a_1\,\psi_1$$

$$\hat{A}\,\psi_2 = a_2\,\psi_2$$

(a) Once the measurement is made and the result is known to be in the state $\rightarrow a_1$

the wave-function collapses to the state ψ_1

(b) Now the state is $\psi_1 \rightarrow$ A measurement of $\hat{B}$ now gives 2 outcomes $\rightarrow$

the particle is in state ϕ_1 with probability $\left|\frac{3}{5}\right|^2 = \frac{9}{25}$

or " " " " " ϕ_2 " " $\left|\frac{4}{5}\right|^2 = \frac{16}{25}$

(c) Note :

$$\left.\begin{aligned} \psi_1 &= \frac{3\phi_1 + 4\phi_2}{5} \\ \psi_2 &= \frac{4\phi_1 - 3\phi_2}{5} \end{aligned}\right\} \rightarrow \quad \frac{4\psi_1 - 3\psi_2}{5} = \frac{12\phi_1 + 16\phi_2 - 12\phi_1 + 9\phi_2}{5} = 5\phi_2$$

$$\left.\begin{aligned} \Rightarrow \phi_2 &= \frac{4}{5}\psi_1 - \frac{3}{5}\psi_2 \\ \text{and } \phi_1 &= \frac{3}{5}\psi_1 + \frac{4}{5}\psi_2 \end{aligned}\right\}$$

A measurement of state :

$$\psi_1 \xrightarrow{\hat{B}} \begin{cases} \xrightarrow{9/25} \phi_1 \xrightarrow{a_1} \frac{9}{25} \equiv \frac{9}{25} \otimes \frac{9}{25} \\ \xrightarrow{16/25} \phi_2 \xrightarrow{a_1} \frac{16}{25} \equiv \frac{16}{25} \times \frac{16}{25} \end{cases}$$

Initial State

$$\text{Total probability} \equiv \frac{81}{(25)^2} + \frac{256}{(25)^2} \equiv \frac{337}{625}$$

3.28.

$$\psi(x,0) = \begin{cases} \dfrac{1}{\sqrt{2n\lambda}} e^{i2\pi x/\lambda} & ; \quad -n\lambda < x < n\lambda \\ 0 & ; \text{ otherwise.} \end{cases}$$

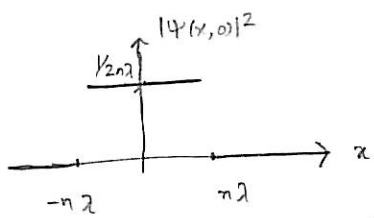
$$\Phi(p,0) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-ipx/\hbar}\, \psi(x,0)\, dx$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int_{-n\lambda}^{n\lambda} \frac{e^{i\left(\frac{2\pi}{\lambda} - \frac{p}{\hbar}\right)x}}{\sqrt{2n\lambda}}\, dx$$

$$= \frac{1}{\sqrt{2\pi\hbar}\sqrt{2n\lambda}} \int_{-n\lambda}^{n\lambda} \left[\cos\left(\frac{2\pi}{\lambda} - \frac{p}{\hbar}\right)x + i \underbrace{\cancel{\sin\left(\frac{2\pi}{\lambda} - \frac{p}{\hbar}\right)x}}_{\text{odd}}\right] dx$$

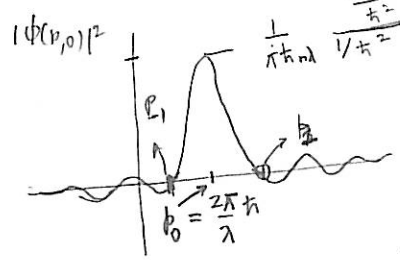
$$= \frac{1}{\sqrt{2\pi\hbar}\sqrt{2n\lambda}} \frac{2\sin\left[\frac{2\pi}{\lambda} - \frac{p}{\hbar}\right]n\lambda}{\left(\frac{2\pi}{\lambda} - \frac{p}{\hbar}\right)} = \frac{2}{\sqrt{4\pi\hbar n\lambda}} \frac{\sin\left[\frac{pn\lambda}{\hbar}\right]}{\left(\frac{p}{\hbar} - \frac{2\pi}{\lambda}\right)}$$

$|\psi(x,0)|^2 \rightarrow$



$$\frac{\frac{n^2\lambda^2}{\hbar^2}}{1/\hbar^2} = \frac{n^2\lambda^2}{\pi\hbar} \leftarrow \text{from L'Hospital's Rule}$$

$|\Phi(p,0)|^2 \rightarrow$



the closest zero's can be found as follows →

$$\sin\left[(p_0 + \delta p)\frac{n\lambda}{\hbar}\right] = 0 = \sin\left[\delta p \frac{n\lambda}{\hbar}\right] = 0$$

$$\Rightarrow \delta p = \pm \frac{\pi}{\left(\frac{n\lambda}{\hbar}\right)}$$

$$\Rightarrow p_1 = \frac{2\pi\hbar}{\lambda} + \frac{\pi}{n}\frac{\hbar}{\lambda} \; ; \; p_{-1} = \frac{2\pi\hbar}{\lambda} - \frac{\pi\hbar}{n\lambda}$$

$$\omega_p \equiv P_1 - P_{-1} = \frac{2\pi\hbar}{n\lambda}$$

$$\omega_x = n\lambda - (-n\lambda) = 2n\lambda$$

$$\Rightarrow \omega_p \omega_x = 4\pi\hbar > \frac{\hbar}{2}$$

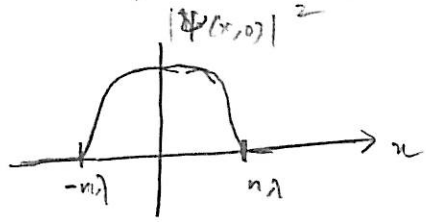
$$\sigma_p \equiv \sqrt{\langle \hat{p}^2 \rangle - \langle p \rangle^2}$$

this term will blow up! Due to the discontinuity in the wave functions at $x = \lambda n$ and $x = -n\lambda$!

Though the uncertainity principle is still valid.

To remedy the problem the wave-function should go to zero smoothly at the edges!!

So $|\Psi(x,0)|^2$ can perhaps have the following form



3.31

$$\frac{d}{dt}\langle x p\rangle = \frac{i}{\hbar}\langle[\hat{H}, x\hat{p}]\rangle = -\frac{i}{\hbar}\langle[\hat{x}\hat{p}, \hat{H}]\rangle$$

$$= -\frac{i}{\hbar}\langle \hat{x}[\hat{p}, \hat{H}] + [\hat{x}, \hat{H}]\hat{p}\rangle$$

Results from the steps given in the solution for 3.17

$$= -\frac{i}{\hbar}\langle \hat{x}\left[-i\hbar\frac{\partial V}{\partial x}\right) + \left(\frac{2i\hbar\hat{p}}{2m}\right)\hat{p}\rangle$$

$$= -\langle x\frac{\partial V}{\partial x}\rangle + 2\langle\frac{\hat{p}^2}{2m}\rangle$$

$$= -\langle x\frac{\partial V}{\partial x}\rangle + \langle 2T\rangle$$

In a stationary state $\rightarrow$ $\psi = \psi_n e^{-iE_nt/\hbar}$ $\leftarrow$ this is a stationary state

$$\langle\hat{x}\hat{p}\rangle = \int \psi_n^* e^{iE_nt/\hbar}\,\hat{x}\,\hat{p}\,\psi_n e^{iE_nt/\hbar}\,dx$$

$$= \int \psi_n^*\,\hat{x}\,\hat{p}\,\psi_n\,dx$$ $\longrightarrow$ No time dependence exponentials cancel out to 1.

thus $\rightarrow$ $\frac{d}{dt}\langle\hat{x}\hat{p}\rangle = 0$

$\Rightarrow$ $\frac{1}{2}\langle x\frac{\partial V}{\partial x}\rangle = \langle T\rangle$ $\longrightarrow$ For stationary state.

$V = \frac{1}{2}m\omega^2x^2 \rightarrow \frac{\partial V}{\partial x} = m\omega^2 x$

$\Rightarrow$ $\frac{1}{2}\langle x\, m\omega^2 x\rangle = \langle T\rangle$

$\Rightarrow$ $\langle V\rangle = \langle T\rangle$